Orchestrating Microbiome Analysis with Bioconductor

Tuomas Borman[1*], Giulio Benedetti[1†], Geraldson Muluh[1†], Aura Raulo[1], Benjamin Valderamma[2], Artur Sannikov[1], Stefanie Peschel[4], Yihan Liu[5], Katariina Pärnänen[1,6], Christian L. Müller[1], Aki Havulinna[1], Sudarshan Shetty, Marcel Ramos[9], Domenick J. Braccia[10], Héctor Corrada Bravo[11], Felix M. Ernst[12], Levi Waldron[13], Thomaz Bastiaanssen[14†], Himel Mallick[1†], Leo Lahti[1*]

[1]Department of Computing, University of Turku.

[1].

[2]Department of Y, Organization XK.

[4], Organization D.

[5]Department of Y, Cornell, Address, City, 0.

[6]Department of XYZ, University of Helsinki.

[9]Department of Y, Organization X, Address, City, 0.

[10]Department of Y, Organization XA, Address, City, 0.

[11]Department of YS, Organization XP, Address, City, 0.

[12]Department of Y, Organization XB, Address, City, 0.

[13]Department of Y, Organization N, Address, City, 0.

[14]Department of Y, Organization XX.

*Corresponding author(s). E-mail(s): tvborm@utu.fi; lemila@utu.fi;

[†]These authors contributed equally to this work.

Abstract

The expansion of microbiome research has led to the accumulation of interlinked datasets encompassing versatile taxonomic and functional assays. While critical to advance the field, the analysis of increasingly large and heterogeneous multi-modal microbiome data would benefit from unified approaches supporting interoperability between methods and the design of modular data science workflows. The Bioconductor project has recently developed an optimized statistical programming framework for multi-assay data integration in computational life sciences. Building on this foundation, we introduce a community-developed open-source ecosystem for microbiome data science, dedicated to supporting joint analysis of hierarchical, interlinked, and heterogeneous datasets that are increasingly common in modern microbiome research. This interoperable infrastructure comprises of open-source datasets, methods, tutorials and an active online community. Its functionality and usage are detailed in the OMA book (https://microbiome.github.io/OMA/docs/devel), which offers guidance for prospective users and contributors.

Keywords: microbiome, Bioconductor, data science, data integration

Introduction

Modern data science applications critically rely on open-source methods created by the research community[1;2] and open software ecosystems have become central innovation hubs, driving technological and cultural transformation[3]. The Bioconductor project has been established as a major distribution channel for open data science methods in the life sciences. Nowadays, its vast developer community maintains over 2,300 quality-controlled R packages for various fields of life science informatics[4]. Additionally, Bioconductor has evolved into a global community that supports data science collaboration and training[5] in general, and now the international calls for enhancing microbiology education in particular[6].

Microbiome research focuses on the study of microbial communities and their diversity, function, and interactions with the host and environment[7]. The technologies used in this field generate vast amounts of high-dimensional, hierarchically structured data. The expansion of microbiome research has led to the accumulation of interlinked, multi-modal datasets encompassing versatile taxonomic and functional assays. Microbiome data scientists are thus increasingly facing the need to integrate heterogeneous data across multiple modalities while implementing reproducible workflows, which presents many practical challenges. First, the unique statistical properties and complexity of microbiome data necessitate specialized approaches[7–9]. Secondly, the diversity of proposed solutions can make it difficult to identify the optimal methods and build interoperable workflows[10;11]. Thus, the need for standardized microbiome data science infrastructures has been widely recognized[7] and a variety of open microbiome data science frameworks have been developed, including Anvi'o[12], Mothur[13], QIIME2[14], and phyloseq[15]. However, platforms that exclusively focus on microbiome data science may not be able to optimally address the need to link data across different multiple modalities originating from different research domains.

The Bioconductor project embeds the methods of microbiome data science into a wider statistical programming framework alongside with the concurrent development of interoperable methods and workflows for transcriptomics[16], single-cell[17], metabolomics[18], proteomics[18;19], and other fields. Here, we introduce a community-developed open-source framework, supporting multi-modal data integration in microbiome research. We have extensively tested and documented the framework to facilitate the dissemination of evidence-based best practices in microbiome data science.

Data infrastructure

Data scientists routinely deal with interlinked assays, side information, and hierarchical data such as ontologies, trees, or networks. The ability to integrate heterogeneous data elements is essential in biological research, where the individual components of a system can be seldom studied in isolation. The challenge for a data scientist is that different methods often require variable and inconsistent input data formats, leading to technical complications and time lost on repeated data wrangling during routine analyses. Bioconductor is renowned for its ecosystem of interoperable statistical methods for life sciences. A key to its broad adoption has been its ability to mitigate those issues by providing standardized data structures supported by an ecosystem of interoperable methods. This helps to avoid wrangling data and aligning methods and leaves more time to be invested in the core analytical tasks.

Bioconductor data structures support statistical analysis of complex data combinations[20] and are widely adopted by various user communities. They have been implemented through specialized object-oriented classes that integrate multiple data elements into a single structure known as a *data container*[21]. The SummarizedExperiment (SE) class represents the primary data container underlying the Bioconductor ecosystem[20;22]. Its position among the top-1% most downloaded packages in Bioconductor reflects its widespread recognition and increasing adoption among developers (Figure 1). SE provides a standardized solution to store tabular data by linking numeric abundance matrices with side information on features (rows) and samples (columns). Side information can include secondary details on the experiment, such as sample origin, collection time, or health status.

The interoperability of SE with other Bioconductor data structures facilitates the transfer of methods across different fields of omics research. This represents a major strength, which has facilitated the development of modular and scalable data science methods[5]. The shared data structures support the transfer of methods between omics fields, yielding improved overall quality control and user support. Consequently, the SE-based framework has been extended to similar data integration tasks in fields, such as single-cell and spatial omics through the SingleCellExperiment (SCE)[17] and SpatialExperiment[16] classes, respectively.

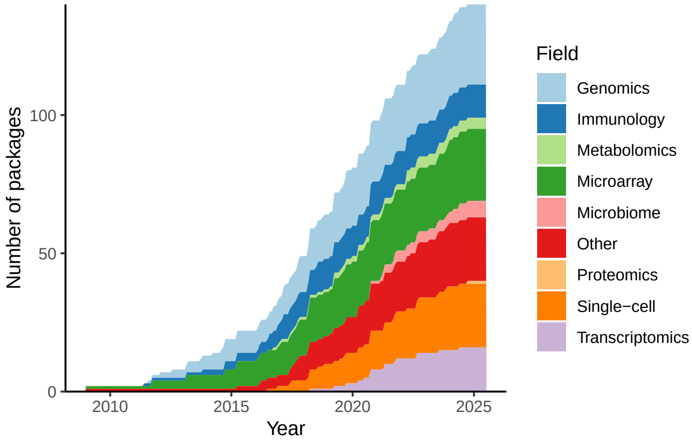
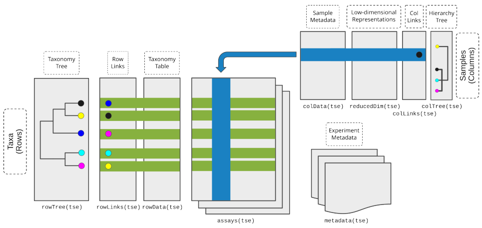


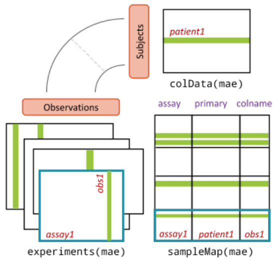
Figure 1: **Adoption of the SummarizedExperiment (SE) data container among Bioconductor developers.** The growth in the number of Bioconductor packages supporting SE over time is shown, categorized by their respective fields. For some of the packages SE support has been added after their original release.

Hierarchical data structures. Microbiome data scientists frequently need to integrate information on feature phylogenies and sample hierarchies in order to accurately reflect fine-scale variability in microbiome data. While maintaining interoperability with the broader Bioconductor ecosystem, the TreeSummarizedExperiment (TreeSE) class[23] extends the SE and SCE data structures to hierarchical datasets by incorporating feature and sample organisations as tree structures (Figure 2). Moreover, TreeSE supports the storage of other common aspects of microbiome data, such as reference sequences, and incorporates results from common analytical procedures such as dimensionality reduction, statistical transformations, and agglomeration by taxonomic, functional or other information. The interoperability with other Bioconductor classes facilitates scalable analysis and the design of modular yet flexible workflows. The direct interoperability with the widely adopted SE data science ecosystem distinguishes our TreeSE-based approach from phyloseq, which is another popular Bioconductor data container originally developed with a focus on 16S amplicon data analysis[15]. TreeSE is a superset that can accommodate all elements of a phyloseq object, while also offering improved support for multi-modal data and often substantial gains in memory efficiency and computing speed (Figure 3). Conversion functions and detailed cheat sheets between the two formats are available to bridge the gap between the respective user communities.

Multi-assay data structures. When integrating data across multiple modalities, such as taxonomy, metabolomics, or host genomics, more flexible sample mapping may become necessary. Multi-omics data is often sparse, and samples can be only partially matched between experiments. In other cases, a single sample in one modality can be matched by two or more samples in another modality see e.g.[26]. The MultiAssayExperiment (MAE) data container[24] facilitates the integration of multiple data objects



(a) TreeSummarizedExperiment



(b) MultiAssayExperiment

Figure 2: ***Bioconductor data containers for microbiome data.*** **(a) TreeSummarizedExperiment (TreeSE).** TreeSE supports hierarchical and multi-table analysis in microbiome data science[23]. It links the numeric abundance matrix and potential alternative versions (statistical transformations, different taxonomic levels, dimensionality reduction) with tabular and hierarchical side information on the features (rows) and samples (columns). Additional slots for reference sequences and experiment metadata are available. (Figure modified from[23])]**(b) MultiAssayExperiment (MAE).** MAE facilitates integration of data from multiple experiments[24]. (Figure from[24])

representing different omics modalities (see Figure 2). Whereas TreeSE is restricted to a one-to-one sample mapping between different data tables, MAE supports flexible mappings between multiple experiments from the same study. Integration of multi-table data without such a structured framework often leads to ambiguous mappings and inconsistent results. For instance, diagonal integration methods[27;28] – where shared features across data types may not be explicitly defined – can benefit substantially from such a systematic framework that enforces anchor alignment. This mirrors lessons from single-cell data integration, where shared anchors and hierarchical mapping have proven essential for interpretability. Flexible linking of samples across several data objects supports efficient data integration, methods sharing, and overall interoperability.

Table 1 summarizes the microbiome-related data containers, while Table 2 introduces the available data elements in each container.

Table 1: Recommended data containers for microbiome analysis.

Data container	Target use
SummarizedExperiment (SE)[20]	generic data container for experiments with multiple assays
TreeSummarizedExperiment (TreeSE)[23]	tailored data container for hierarchically organized experiments
MultiAssayExperiment (MAE)[24]	tailored data container for sets of multi-omic experiments

Table 2: Key data elements in the employed data containers.

Slot	Content	SE	TreeSE	MAE[a]
Tables				
assays	tables of features (rows) and samples (cols)	X	X	X
altExps[b]	experiments with variable number of features		X	X
reducedDims	results of dimensionality reduction		X	X
Rows				
rowData	side information on features	X	X	X
rowPairs	linkage between pairs of associated features		X	X
rowTrees	hierarchical organization of features		X	X
referenceSeq	reference sequences of features		X	X
Columns				
colData	side information on samples	X	X	X
colTrees	hierarchical organization of samples		X	X
sampleMap[c]	flexible sample mapping across experiments			X
Other				

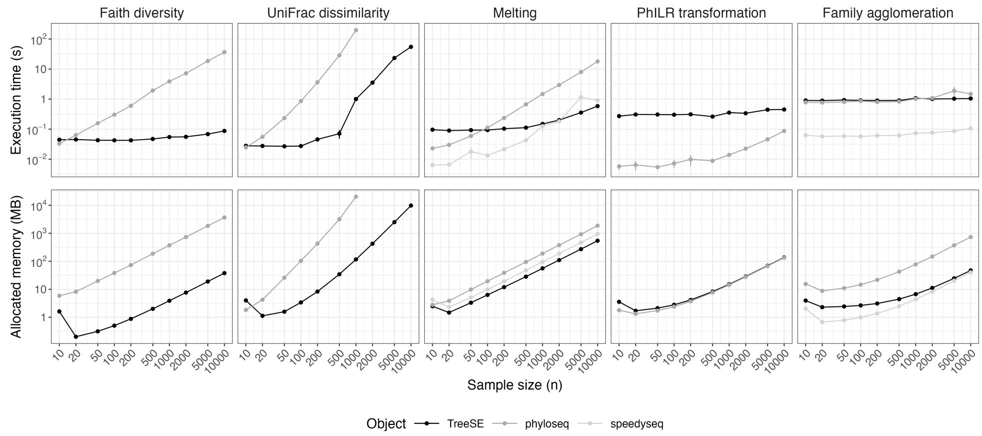


Figure 3: **Benchmarking mia/TreeSE against phyloseq and speedyseq.** TreeSE, phyloseq and speedyseq were benchmarked on five common analytical operations using random subsets of samples from the GrieneisenTSData experiment[25]. Execution time (s) values and error bars represent the mean and standard error (SE) over 10 iterations, respectively. Allocated memory (MB) was estimated based on a single iteration. Overall, TreeSE outperformed phyloseq. While speedyseq achieved higher performance in the supported operations, TreeSE scaled more efficiently in both execution time and memory allocation with increasing sample size (n). The benchmarking analysis was conducted in R using the bench library with 8 CPU cores and 32 GB RAM. All operations, except for UniFrac dissimilarity with phyloseq, could also be performed on a regular laptop (e.g., 4 CPU cores and 16 GB RAM).

[a] While MAE does not contain conventional slots (with the exception of colData), its experiments can include any number of data containers that provide those slots. [b] altExp samples must match assay samples in a 1-to-1 mapping. [c] sampleMap supports diverse mappings between samples from different experiments (1-to-1, 1-to-many, many-to-1 and many-to-many).

Data resources. Several open data resources support this framework for microbiome analysis and direct data import into the aforementioned data containers for further downstream analysis in Bioconductor. Examples of the available data resources include curatedMetagenomicData[29], HoloFood[30], microbiomeDataSets[31], and the EBI/MGnify database[32]. In addition, Bioconductor packages often include built-in demonstration datasets. Collectively, these resources encompass taxonomic and functional profiles from thousands of published microbiome studies and tens of thousands samples that can be accessed with the available tools for research and educational purposes (see Table 3).

Table 3: Open microbiome data resources supporting the framework.

Data resource	Package	# studies or datasets	# samples
Curated metagenomic data [29]	curatedMetagenomicData [29]	93	22,588
HoloFood [30]	HoloFoodR [33]	-	9,990
MGnify [32]	MGnifyR [34]	5129	616,138
microbiomeDataSets [31]	microbiomeDataSets [31]	6	19,100

Data preparation

While microbiome data scientists continue to develop specialized solutions [8;9;35], the implementations typically differ in their expected input and output formats. Common data standards support modular data science workflows, speeding up methods development, benchmarking, and replication. Here, we describe an ecosystem of software packages that has recently emerged to support microbiome data integration and analysis based on the data containers, (Tree)SE and MAE (see Figure 4).

The framework focuses primarily on the downstream analysis of high-throughput sequencing data from microbiome experiments after the original measurements have already been summarized into abundance tables representing e.g. taxonomic or functional profiles. Taxonomic abundance tables are one of the most commonly encountered data types. Common measurement platforms include amplicon-based marker gene studies (e.g. 16S rRNA) and metagenomic sequencing, but alternative profiling techniques are available including phylogenetic microarrays [36] and high-density optical mapping [37]. Microbiome analyses can also yield variable metagenomic, and functional features. The interoperability of the data formats supports modular workflow design for integrating different omics through the Bioconductor ecosystem. This decisively mitigates reproducibility challenges posed by rapidly evolving reference databases and profiling tools (e.g., MetaPhlAn2 vs. MetaPhlAn4) by enabling structured coexistence and direct comparison within a unified, version-resilient framework [38;39].

Data import. Most users will need to import data for their own analyses. The TreeSE data container can be populated with microbiome data from common tabular data types but non-standard file formats may require additional data wrangling. To streamline data import, we provide importers for a variety of standard microbiome file formats including BIOM [40], MetaPhlAn/HUMAnN (MetaPhlAn v. 2-4, HUMAnN v. 3) [41], Mothur [13], QIIME2 [14] and the TAXonomic Profile Aggregation and STAndardisation format (taxpasta) [42]. These importers can automatically load the data into the appropriate data containers, thus forming a bridge between low-level sequence processing methods and the high-level statistical programming framework in R. Moreover, converters between alternative data structures in R are available supporting seamless conversions between TreeSE and BIOM, DADA2 [43] and phyloseq (the mia::convert* functions), and TreeSE-based microbiome data objects can be exported into external services such as MicrobiomeDB[44] and in other languages (e.g. Julia). These bridges facilitate the broadest possible access to microbiome analysis methods.

Quality control and exploration. After importing the data into R, the recommended first step involves basic data exploration and quality control[45]. In addition to methods for identifying contaminant sequences and calculating summary statistics, the framework includes a versatile set of custom visualization methods to assess data variability, outliers, homogeneity of variance and other aspects that may need to be considered in downstream data analysis. The mia and miaViz packages, for instance, provide convenient access to many operations for common exploratory tasks in microbiome analysis and visualization for TreeSE objects (Figure 5).

Filtering and agglomeration. Microbiome datasets may contain tens of thousands of unique features depending on the sequencing resolution, databases, and the scope of analysis. Often – to summarize data into biologically meaningful groups or to reduce multiple comparisons – the data is subsetted, filtered or agglomerated to higher-level taxonomic or functional categories or stratified by sample groups. The package ecosystem and the mia package feature various tools for such tasks. For instance, the prevalence methods can be useful to remove very rare features that might be sequencing artifacts or present otherwise limited value for the analysis. The mechanism of alternative experiments (altExp) supports data agglomeration by taxonomic levels and by other features such as pathogeneicity or functional potential (e.g. butyrate production[46], gut-brain modules[47], and other host-associated signatures[48]). The supporting methods incorporate the agglomerated levels within the same data object for parallel operations, and typically also take care of agglomerating the associated tree information and the alternative assays. The filtering, subsetting, and agglomeration operations are thus supported in both feature as well as sample spaces, and simultaneously propagate to cover the relevant alternative assays, alternative experiments, ordinations, and side information. This supports the efficient management of interlinked assays and saves time for dedicated expert tasks.

Transformation. Data transformations are frequently encountered in statistical analysis. Common transformations in microbiome research include relative abundance[49], centered log-ratio (CLR) and its robust version (rCLR), which are used to mitigate compositionality bias[50], and the tree-aware phylogenetic ILR (phILR) transformation[51]. Data rarification is a related concept that has been suggested to control for uneven sequencing depths and it remains widely debated in microbial ecology[52;53]. Our framework supports these and a variety of other transformations commonly applied to microbiome data.

Data enrichment. Finally, incorporating additional data is a frequent operation in applied data analysis, and it often takes place in an iterative fashion. The data containers support gradual and organized accumulation of interlinked data elements including transformations, aggregated data, dimension reduction, and derived data on samples and features. We generally recommend that the data objects are initialized at the beginning of the workflow, where they can then be gradually enriched as the project advances. By systematically extending the upstream data preparation step –

instead of ad-hoc calculations scattered along the workflow – the data scientist can achieve well-organized and transparent provenance of the data elements. Data can be enriched by importing new external information as well as by deriving new information from the data itself into the same data container as described in the next section.

Downstream data analysis

The new analysis framework improves the management of analysis workflows by systematically linking the different data elements and analysis outcomes. The framework is thus shifting the focus towards multi-modal analyses and methods development while at the same time supporting the essential standard techniques in microbiome research, such as community diversity and composition. Microbiome data analysis often relies on specialized statistical approaches as data sparsity, zero-inflation, compositionality, and heteroschedasticity pose challenges for standard data analytical methods[8;49]. Moreover, microbiome data is hierarchically structured across taxonomic, ecological, and spatial levels[7;54]. Alternative processing pipelines pose different biases and generate varying data formats[55]. These and other unique characteristics have wide-reaching implications for downstream analysis, necessating dedicated processing and analysis methods in microbiome research[7]. Our statistical programming framework supports the community-driven development and benchmarking of these solutions. The following sections provide an overview of the available tools for microbiome data science.

Community indicators. Various global indicators are available to quantify the microbial community state or other sample properties. Community diversity represents a common measure in microbial ecology[56] (Figure 5). Alpha diversity metrics generally quantify the observed or estimated number of distinct species ("richness") and their distribution ("evenness"). Many diversity indices, such as the Shannon index, combine these two aspects. Phylogenetic diversity measures additionally consider the phylogenetic relatedness of community members[57]; the new implementation of the Stacked Faith's Phylogenetic Diversity can accelerate the performance 1-2 orders of magnitude[58]. Enhanced analyses with repeated rarefaction are supported[59] albeit the use of rarefaction remains controversial[53;60]. Other global indices of community state include measures of dominance, rarity, skewness, kurtosis, modality, library size (number of reads), or absolute abundance quantification. Sample information in the TreeSE object (colData) can be enriched with these derived quantities. By default, the mia::addAlpha method returns a set of complementary alpha diversity measures[61].

Community similarity is commonly evaluated with beta diversity measures, which quantify multivariate dissimilarity and can be used to quantitatively compare community composition. The framework supports common indices in microbial ecology including Bray-Curtis, Jaccard, Aitchison[50;56], and the phylogeny-aware UniFrac[62]. The community (dis)similarities are commonly visualized with unsupervised ordination methods that project the data on a lower-dimensional display[63]. Unsupervised techniques, such as Principal Component Analysis (PCA) and Principal Coordinates Analysis (PCoA) also known as Multi-Dimensional Scaling (MDS), and Non-metric

MDS (NMDS) (Figure 5) are common choices, with enhanced support for microbiome analysis and visualization. Statistical tests (e.g. PERMANOVA) and supervised visualization methods, such as the distance-based redundancy analysis (dbRDA) can complement the unsupervised analyses, and help to uncover and quantify co-variation between community composition and covariates. This can benefit from compositionally robust measures (e.g. robust Aitchison distance[50]), or phylogeny-aware dissimilarity measures specifically developed for microbiome research (e.g. Unifrac[62]). Sample dissimilarities provide the basis for many specialized microbiome techniques such as microbial community typing (clustering) with Dirichlet Multinomial Mixtures (DMM)[64] and community state types (CST)[65], or factorization into community signatures[66]. These and other multivariate methods are provided for instance by mia, miaViz, and additional single-cell packages (scater, scuttle[17]).

Differential abundance analysis (DAA). DAA methods are used to identify microbial taxa whose abundances vary across study groups or continuous gradients (Figure 5). Many dedicated DAA methods in microbiome research support the TreeSE framework including methods such as ANCOM-BC2[67], LimROTS[19], MaAsLin3[68], and radEmu[69]. Recent studies by us and others indicated that elementary methods, such as log-transformed TSS-counts coupled with a linear model, or presence/absence data with logistic regression often yield the most consistent results[70]. The standard DAA typically focuses on individual features, ignoring interactions and potential co-abundance variation. Thus, we support co-abundance analyses; agglomerating collinear features as a preprocessing step can reduce multiple testing, increase statistical power, and support interpretation but is also prone to subjective parameterization and limited generalizability. The grouping can be based on clustering of the abundance profiles or on external information such as phylogenetic relations, taxonomic classifications or functional properties (e.g. butyrate producers[46]). Beyond these, advanced methods—such as causal mediation analysis tailored to microbiome multi-omics data—are already being developed using SummarizedExperiment[71], highlighting its readiness to support emerging innovations in microbiome analytics.

Function- and sequence-level analyses. Taxonomic analysis is frequently complemented by functional analyses, either based on functional predictions derived from amplicon or metagenomic sequences, or from parallel metabolomic or other functional assays. Our framework not only supports incorporation of functional abundance tables within the same data objects but specialized methods on functional data are available; for instance, regarding metabolome analyses the Notame package[72] recently added SE support, and the R4MS project provides many interoperable tools[18]. The framework has found recent applications in extended analyses of metagenomic features, i.e. antimicrobial resistance gene composition, diversity, and load[73]. We have converted a large-scale compilation of resistome profiles in 8,972 metagenome samples[74] in the TreeSE format for research and educational purposes.

Spatio-temporal analyses and ecological modeling. Microbiome time series[75] encompass a wide range of study designs, ranging from short to long follow-up times, sparse to dense data, univariate to multivariate and multi-omic monitoring[76], and real, pseudo, or simulated time series. The miaTime R package provides automated solutions for longitudinal microbiome analyses. Moreover, an increasing number of studies have incorporated the spatial dimension in microbiome analysis across scales[54]. While spatial and temporal analyses require specialized methods, solutions in Bioconductor are available from other application domains, such as single-cell sequencing[17] and spatial transcriptomics [**(author?)** [16]}, which use closely related data containers (Figure 5). Spatial and temporal information can be incorporated as a covariate or by estimating temporally or spatially resolved metrics, such as divergence, auto-correlation and modality. Another type of time series analysis includes the time-to-event, or survival models, which are encountered in e.g. population cohort studies; specialized survival analysis strategies for microbiome data have been proposed[77;78]. We also provide methods to simulate time series from standard models of microbial community dynamics including e.g. the Hubbell's neutral model, generalized Lotka-Volterra, and consumer-resource model through the miaSim package[79].

Multi-modal integration. An increasing number of studies integrate microbial abundance data with further omics, such as transcriptomics, proteomics, and host genomics[7;80–82] (Figure 5). As demonstrated above, Bioconductor data containers support combinations of complex omics data.[17;20;24]. Statistical methods for multi-table analysis in microbiome research can be broadly categorized into three groups: Machine Learning (ML) predictive modelling[83], association-based modelling, and latent variable modelling.[7;8;83–86]. Some of the currently supported multi-omic methods include cross-correlation analyses, canonical correspondence analysis (CCA), anansi[85], and joint-RPCA [in press].

Other methods and programming environments. A range of other methods are employed in microbiome data science. Examples include network analysis[87], ML-based prediction and classification[88], enrichment analyses such as BugSigDB[48] or Microbe Set Enrichment Analysis (MSEA)[89]. The recent adoption of of interoperable data containers and the methodological similarities in related fields can greatly expand the selection of available toolkits in emerging areas. In addition to the internal coherence within the Bioconductor ecosystem, this framework supports integration with external platforms through interfaces such as `reticulate` for Python, `Rcpp` for C++, and compatibility with pipelines like bioBakery[41;90]. These connections empower users to combine the best tools available across environments and languages while maintaining structured data interoperability via standard containers like TreeSE and MAE.

Accessible analysis and community

Modern science is embodied by a critical community capable of assessing discoveries and replicating results[91]. Open software and the social structures around it present a timely manifestation of this principle, with technical, legal, sociological and epistemic consequences[3]. FAIR principles for software support access to well-documented and user-friendly methods[1], and are facilitated by the extensive quality control and documentation standards enforced by Bioconductor[20]. In addition to these technical demands, the development and quality control of academic software critically relies on active critical source communities that can provide continuous benchmarking and support the identification and dissemination of evidence-based best practices[1;3;75]. We actively support community formation by online training resources and communication channels. These, together with the interactive applications, provide accessible entry points for prospective developers and users.

The OMA book. *Orchestrating Microbiome Analysis with Bioconductor (OMA)* (https://microbiome.github.io/OMA/docs/devel/) is an online book that provides a comprehensive introduction to the microbiome data science framework in Bioconductor. It offers practical guidance, reproducible workflows, and training material for integrative microbiome analyses. The book is versioned to match Bioconductor's biannual software release cycles. The work incorporates rich feedback from the user community and microbiome training courses. It complements the broader trend, where specific Bioconductor data structures[20;23;24] have been adopted as a solution for multi-table data integration tasks in various domains such as single-cell sequencing (OSCA book)[17] and transciptomics OSTA book. These resources demonstrate how state-of-the-art data containers can support statistical programming on interlinked multi-table data sets in a highly optimized way.

Interactive applications. Graphical user interfaces provide accessible entry points to the framework. The iSEEtree and miaDash packages provide interactive analysis and visualization tools for microbiome data[92] (Figure 5). Their interface supports automatic generation of source code and replicable workflows as well as analysis templates for users without a strong programming expertise. Moreover, the tidy R paradigm, available through tidyomics, simplifies the syntax and can streamline the workflows[93]. These tools allow users to perform basic analyses and visualize data with ease.

Community. Its widespread adoption has made R one of the most cited software resources of our time[2] and the introduction of the shared data structures facilitates the broad interoperability of the methods ecosystem. Bioconductor has a strong tradition in fostering the R user community through shared data structures and training activities[5]. An active community of developers and users thrives online, supporting the software sustainability and promoting good practices in scientific computing[94]. The community interacts via several online communication channels, including Bioconductor Zulip, a Github forum, and an email list. The development of the framework is influenced by feedback from the user community through these channels, regular meetings and training workshops.

Outlook

The Bioconductor community has a well-established reputation in developing open, peer-reviewed software for life sciences. Its strengths include the enforcement of minimal documentation standards, semantic versioning, portability across operating platforms, quality control (e.g. unit testing, continuous integration), usability (APIs), and robustness (deprecation mechanisms), and a biannual release cycle. During the past two decades Bioconductor has grown into a repository of 2,300 actively maintained software packages, of which 53 are classified under the Microbiome Bioc task view. In order to enjoy the benefits of automation while maintaining flexibility, data scientists routinely remix open-source software into custom workflows. However, the lack of commonly adopted data representation standards often hinders the development, sharing, and application of statistical workflows. At the core of the Bioconductor's data science strategy is the idea of shared data representation. The enhanced interoperability can help to minimize redundant efforts and facilitate the translation of methods between different research areas[1;95] and, importantly, facilitates the formation of broad developer and user communities.

Microbiome research has shifted towards integrative analyses aiming at mechanistic and causal inference[56;96]. Technological advances in long-read sequencing, functional assays, spatial profiling, and single-cell have created an increasing demand for integrative data analysis techniques. Thus, microbiome data is increasingly linked with other omics along varying spatial and temporal dimensions. Our framework addresses this need, together with parallel developments in other areas of omics.

Recent developments in microbiome and multi-omics workflows increasingly rely on Bayesian modeling and Gaussian processes to accommodate uncertainty and temporal or spatial structures[97]. Our framework is compatible with these approaches, providing the structured input necessary for probabilistic inference and scalable computation. Building on these statistical foundations, the rapid emergence of foundation models and deep learning approaches in biomedicine calls for robust, scalable data representations[98]. Our framework, through its use of standardized containers and modular workflows, is well-positioned to support such advances. Applications include transfer learning across studies, interpretable embeddings of microbiome multi-omic profiles, computational prediction of microbial function and metabolites[99;100], and integration with AI agents for hypothesis generation and automated analysis, among others[101;102].

Open data science infrastructures have become critical components of reproducible research and a vast body of applications critically rely on open-source methods and workflows[1;3;75]. Standardized data structures can enhance the transparency, reliability, and explainability of analysis workflows. We have documented the interoperable framework built on widely adopted data containers, open data resources, analysis methods and reproducible workflows for multi-modal microbiome research. Further developments could include extended methodology for temporal, spatial, and multi-omic analyses. Supporting interoperability between R and other programming languages can facilitate the application of other ML frameworks. As the field evolves, Bioconductor will continue to integrate new tools and approaches to improve integrative analyses of microbiome data.

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Consortia

The mia contributors: **we will list here main and consortium authors who approved their inclusion** on this sheet. Sorted in alphabetical order by last name.

Contributions

LL and TB initiated and coordinated the work. All authors contributed material, participated in manuscript preparation, and approved the final version.

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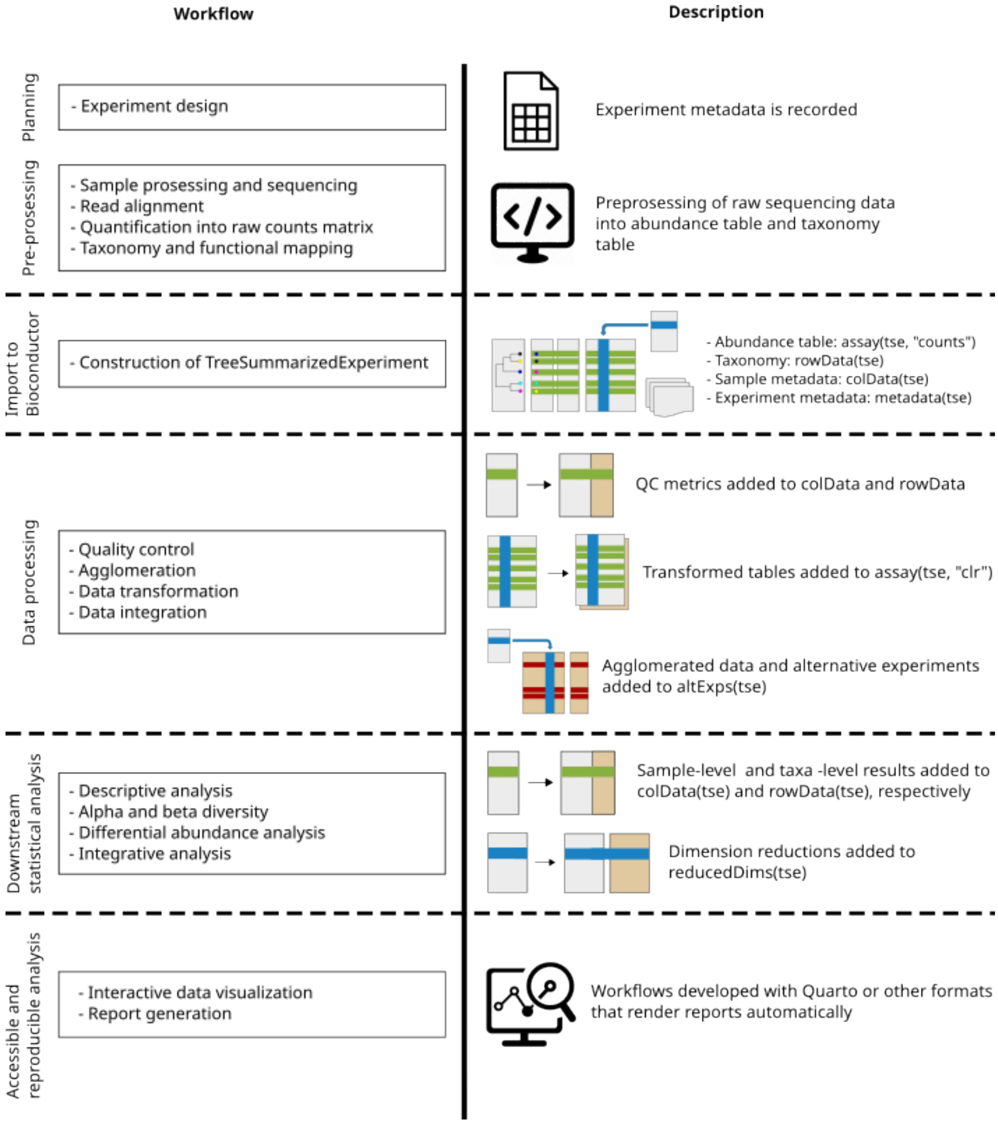

Figure 4: **Examples of Bioconductor tools along the microbiome data science workflow.** A typical microbiome workflow starts with taxonomic and functional annotation. The processed microbiome data can then be imported to Bioconductor data container (TreeSE), or integrated with other omics using the MAE container for downstream statistical analysis. (Figure adapted from[17])

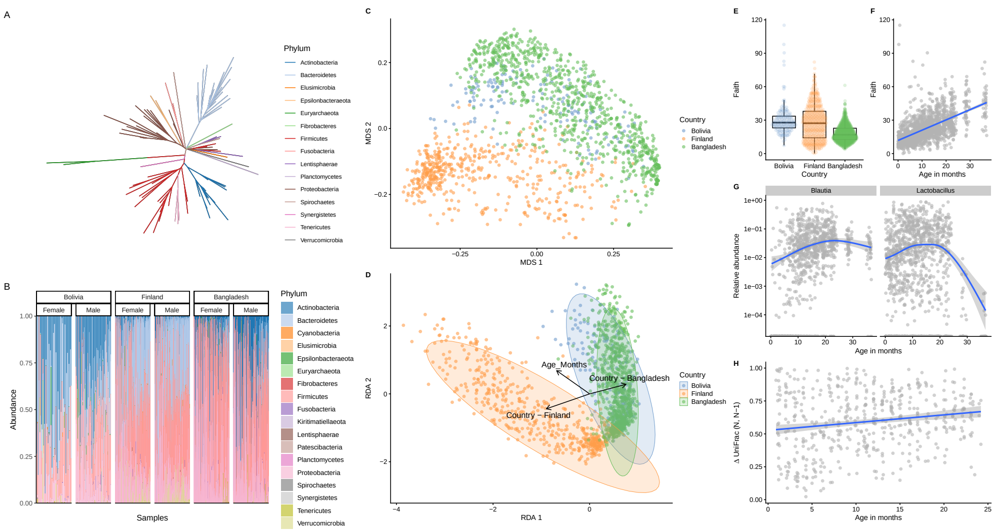

Figure 5: **Common visualization methods.** This set of visualizations present time series data on infants from three different countries. **Figure A** illustrates the phylogenetic relationships, while **Figure B** summarizes phyla abundances in the samples ordered based on age. Actinobacteria is more present in Bolivia and Bangladesh compared to Finland. **Figures C and D** display the results of Multidimensional Scaling (MDS) using Unifrac distances and Distance-based Redundancy Analysis (dbRDA) based on Aitchison distance, respectively. Both ordination methods shows clear pattern separating microbial profiles in different countries. dbRDA shows additionally association between microbial profile and age. **Figures E and F** depict Faith's phylogenetic diversity visualized in respect to country and age. **Figure G** visualizes Differential Abundance Analysis (DAA) results with scatter plot. Two genera exhibit association with age. **Figure H** illustrates Unifrac divergence for consecutive time points. Results indicate slow stabilization of microbial communities.